

A biologically informed AI framework for stratifying cancer patients

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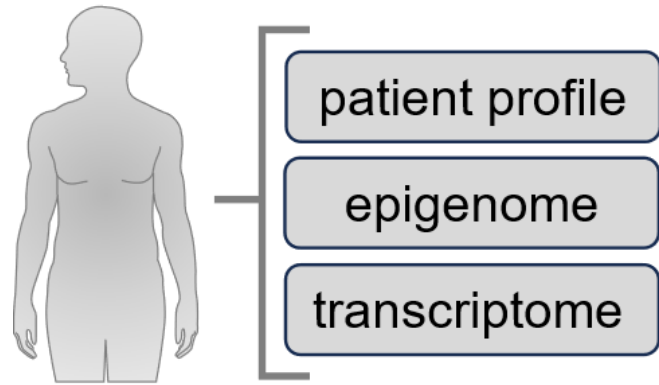
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Motivation

- Cancer is a heterogenous disease with vast genomic variations that can affect patients' survival time.
- Conventionally, cancer patients are classified (graded) based on histological images to predict their prognosis.
- With the available of multi-modal molecular data, we investigate the association between patients' survival time and their genomic variation, aiming for refining patient classification.

Overview of the approach

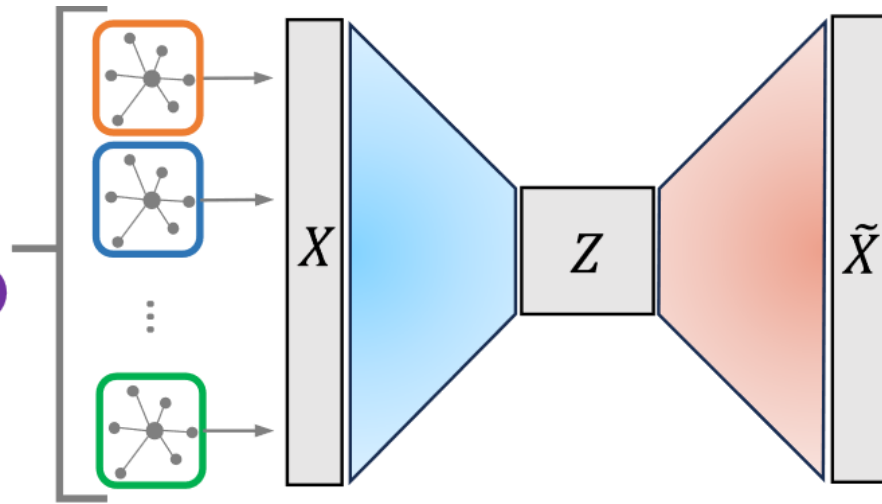
multimodal data



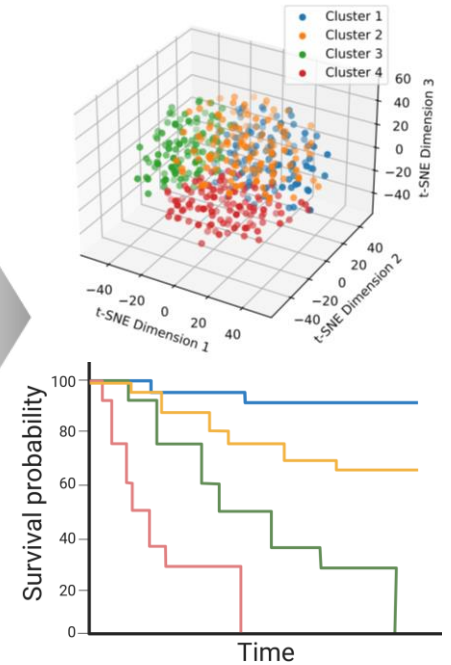
disease network



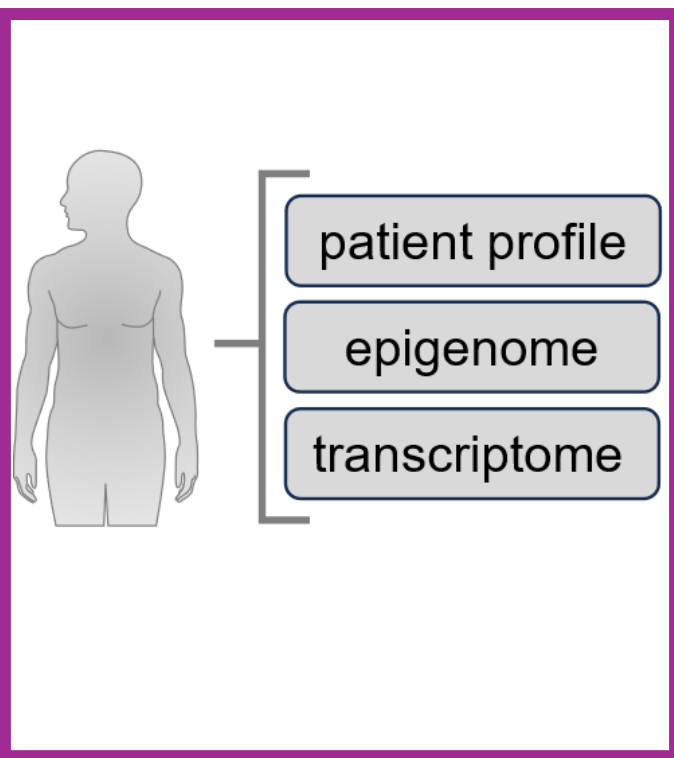
artificial neural network



patient clustering



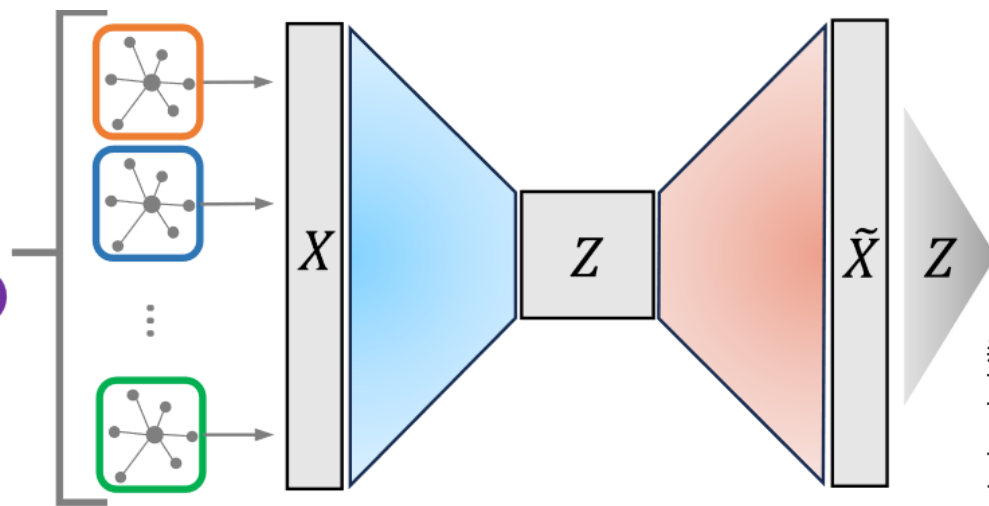
multimodal data



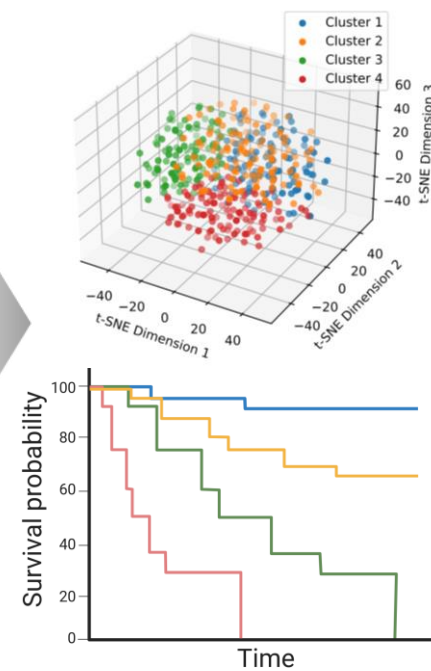
disease network



artificial neural network



patient clustering



Overview of the TCGA genomic and clinical data of cancer patients

Cancer	Patient	Cancer	Patient
BLCA	397	SKCM	464
BRCA	658	UCEC	377
COAD	294	TGCT	145
ESCA	171	MESO	77
KICH	65	UVM	79
KIRC	267	HNSC	495
KIRP	257	UCS	55
LIHC	357	CHOL	34
LUAD	451	CESC	279
LUSC	356	ACC	76
PAAD	155	PCPG	173
PRAD	477	DLBC	47
READ	93	LGG	517
THYM	109	SARC	231
STAD	357	GBM	54
THCA	404		

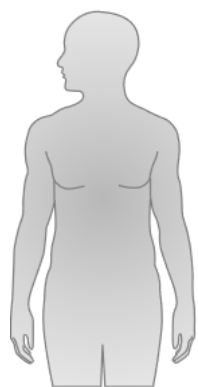
SKCM patient demographic data

Age	Gender	Tumor type	Tumor stage	Radiation	Therapy
(mean $\pm$ s.d)	(male/female)	(metastasis/primary)	(0/i/ii/iii/iv)	(yes/no)	(yes/no)
58.21 $\pm$ 15.79	288/176	360/104	7/75/139/168/23	114/350	145/319

SKCM genomic data

Category	Value and range	Processed	
		Identifier	No. of features
Transcriptome (gene expression)	FPKM $\mathbb{R} \geq 0$	Gene symbol	55 500
DNA methylation	Beta value $\mathbb{R} \in [0, 1]$	Gene symbol	33 487

multimodal data

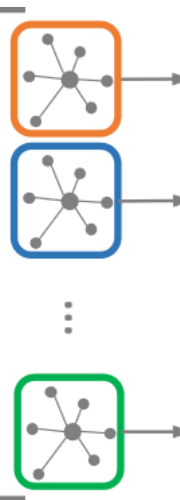


patient profile

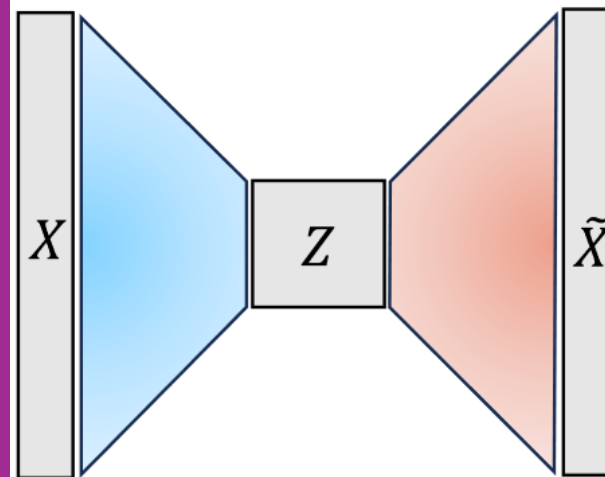
epigenome

transcriptome

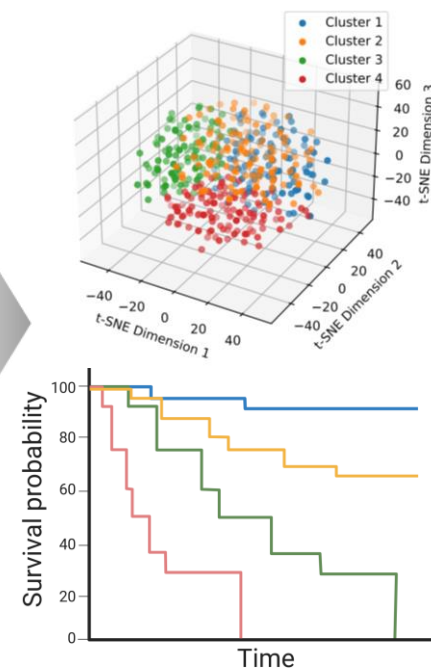
disease network



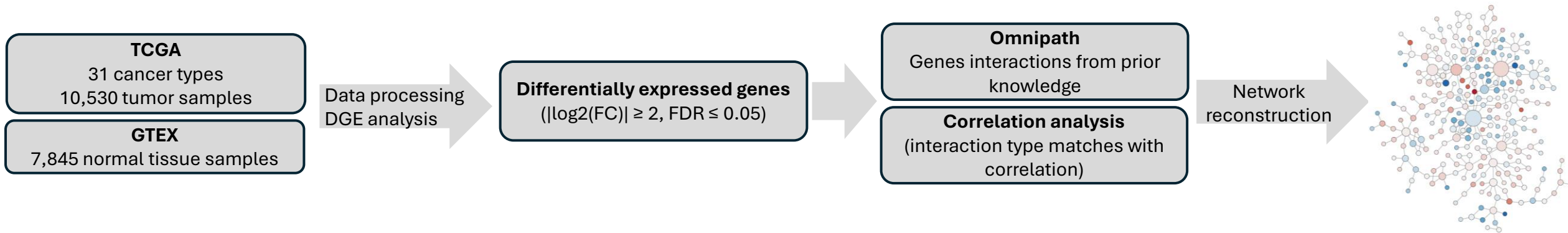
artificial neural network



patient clustering



Reconstruction of cancer-specific networks



31 cancer networks

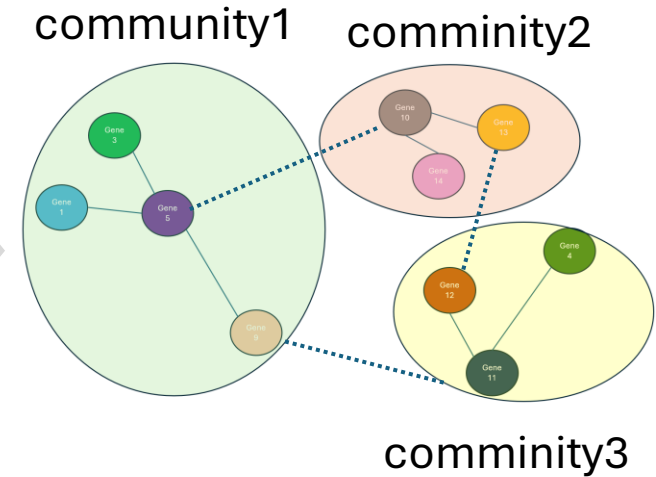
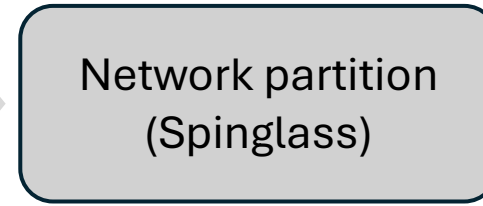
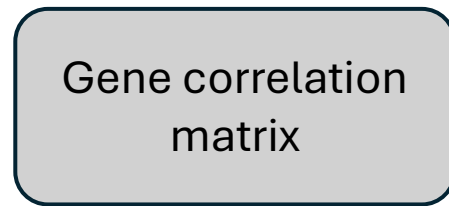
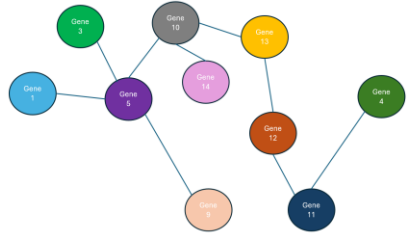
Cancer_Type	Nodes	Edges	Cancer_Type	Nodes	Edges	Cancer_Type	Nodes	Edges
BLCA	4797	25556	LUSC	8245	50843	ACC	6557	35877
BRCA	7903	48254	PAAD	7509	45982	CHOL	6729	39772
COAD	7888	48251	PRAD	7488	44292	CESC	3917	18270
ESCA	7449	42951	READ	7446	44847	GBM	7831	47502
KICH	5684	28217	SKCM	8247	51167	HNSC	7728	47290
KIRC	6493	36709	STAD	7520	45904	DLBC	7051	40318
KIRP	5672	30433	THCA	7945	48464	LGG	7733	46838
LIHC	6796	40344	THYM	8109	50992	MESO	7705	45095
LUAD	7981	47806	UCEC	7345	44274	UVM	7888	48559
PCPG	7444	43984	SARC	8056	49865	TGCT	8195	50185
UCS	7292	41827						

DGE: Differential gene expression

Omnipath: a database for experimentally validated molecular interactions

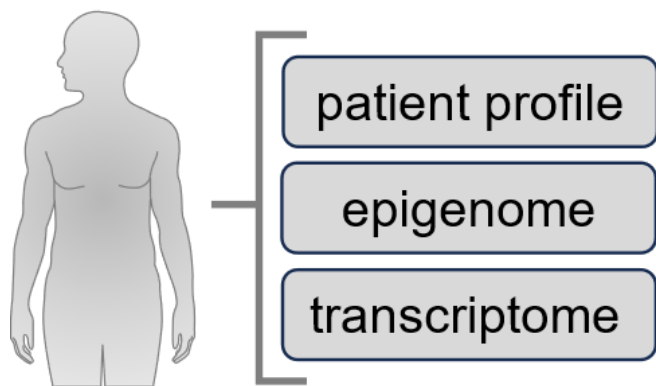
Identification of network community

Cancer-specific network



- Singlass algorithm is a simulated annealing algorithm to identify network communities.
- Network community detection is used to find subnetworks with high cohesion and low coupling.
- Network community can uncover and interpret functional clusters within complex biological systems.

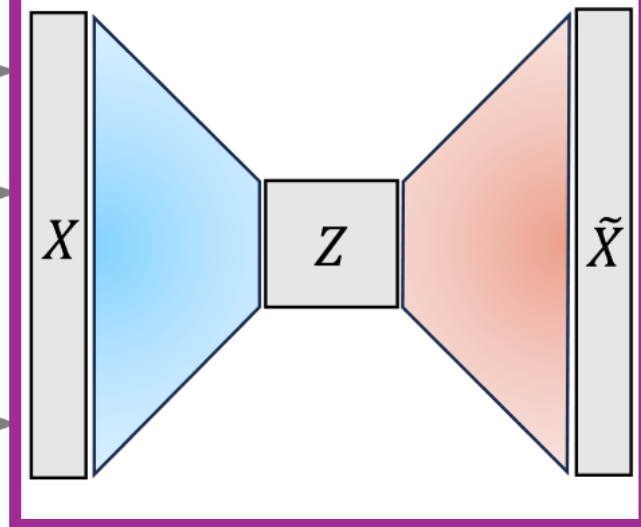
multimodal data



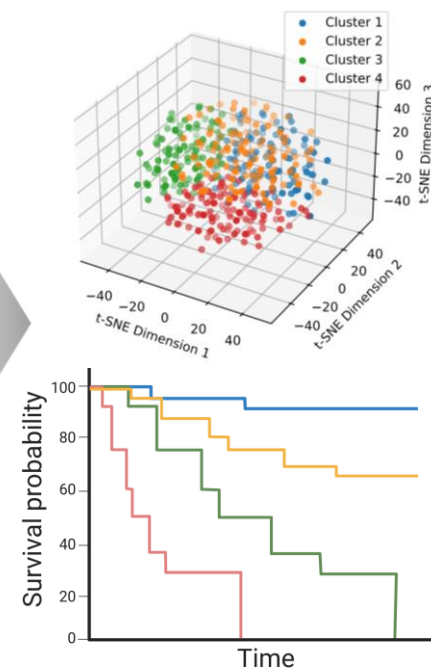
disease network



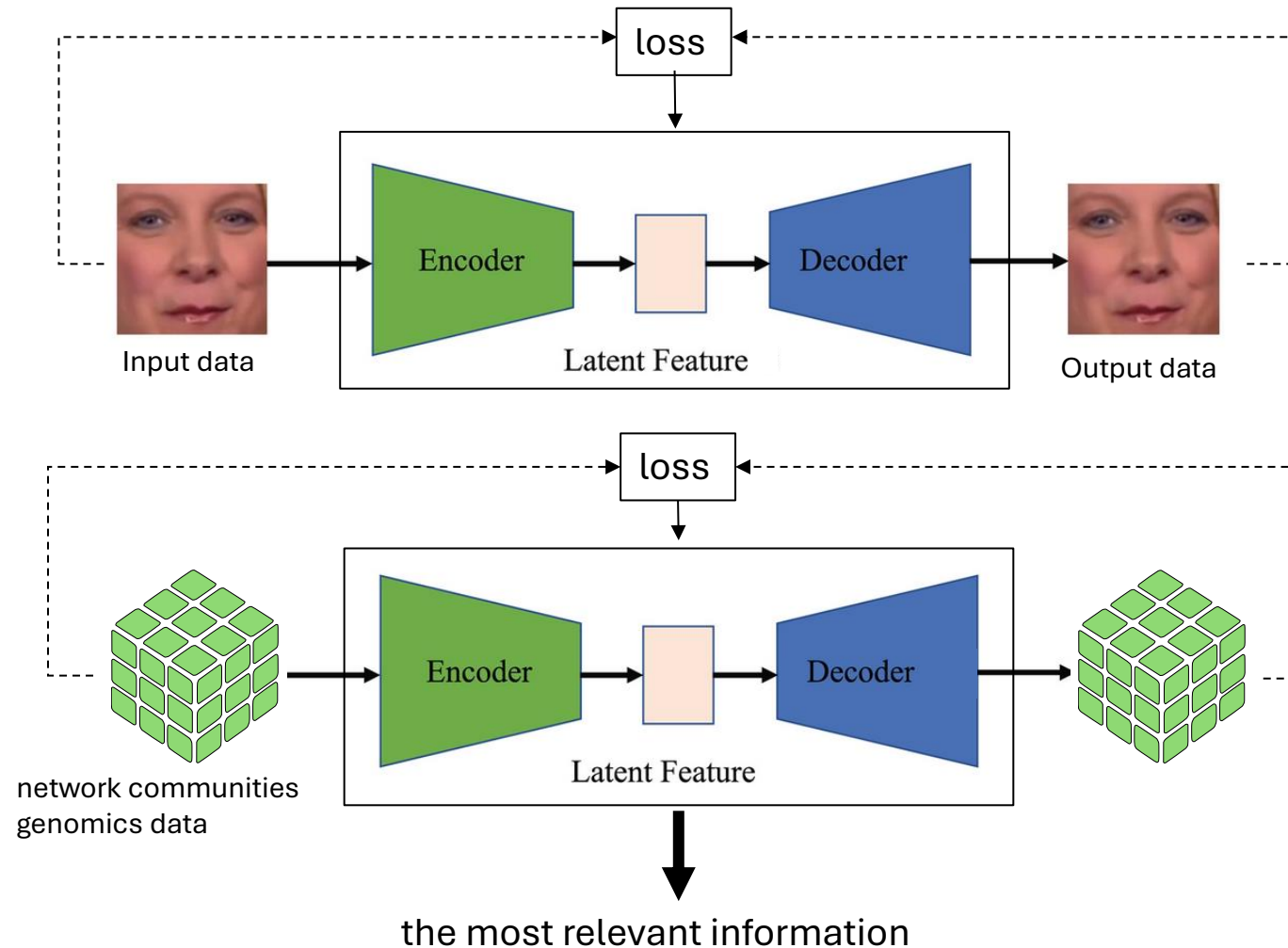
artificial neural network



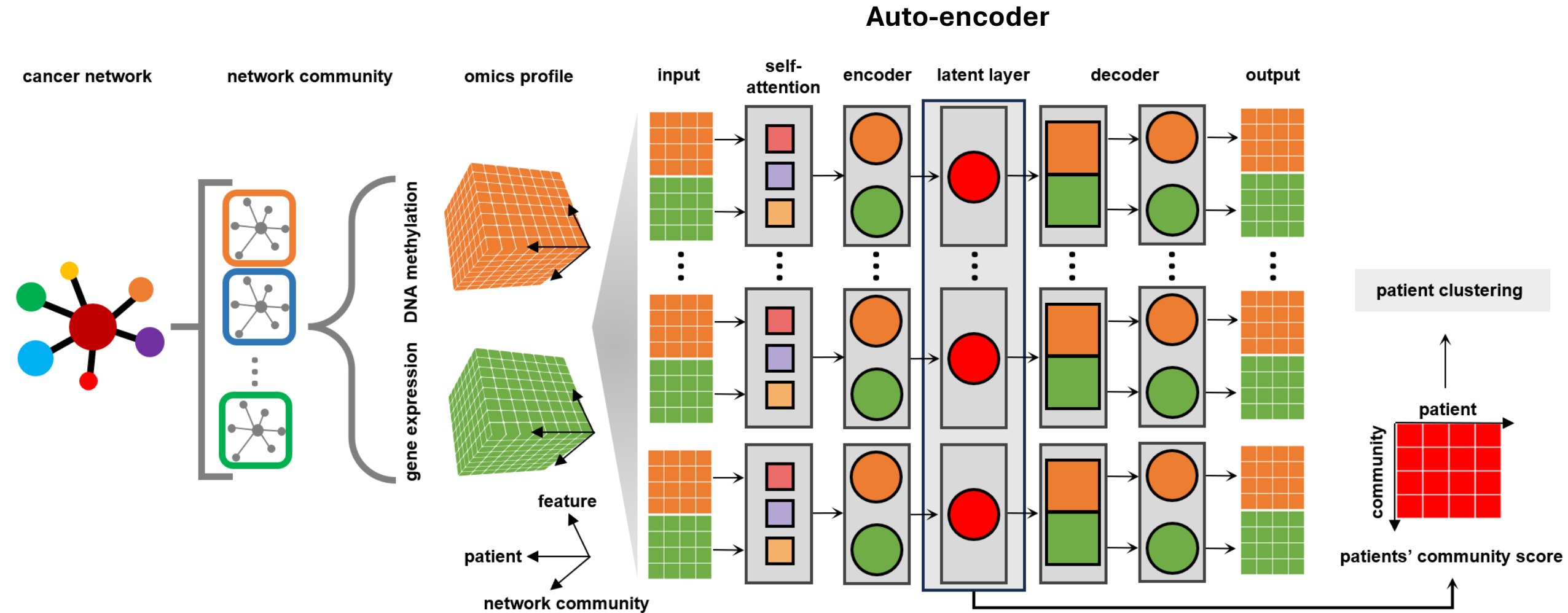
patient clustering



An autoencoder model to reduce the dimension of the genomics data

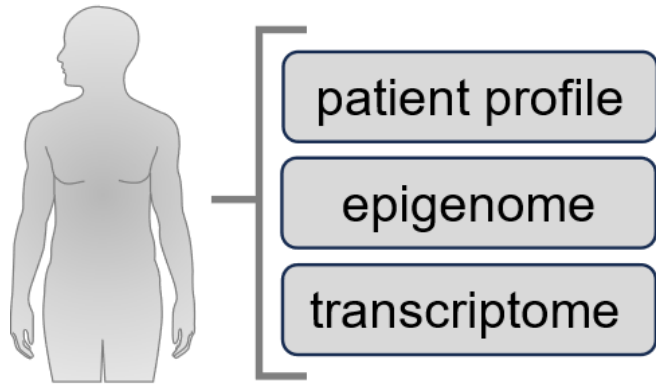


The cancer network-informed autoencoder



We used parallel computing to reduce the computation time from **363 hours** to **2 hours** for 31 cancer types.

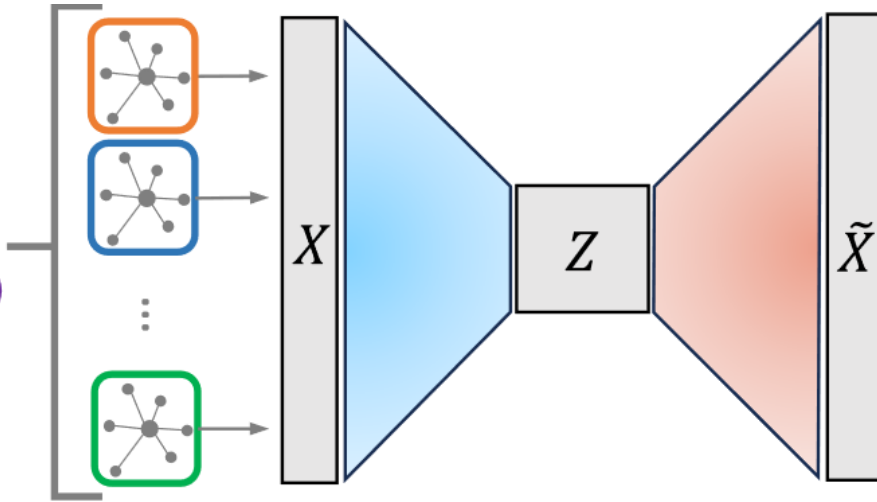
multimodal data



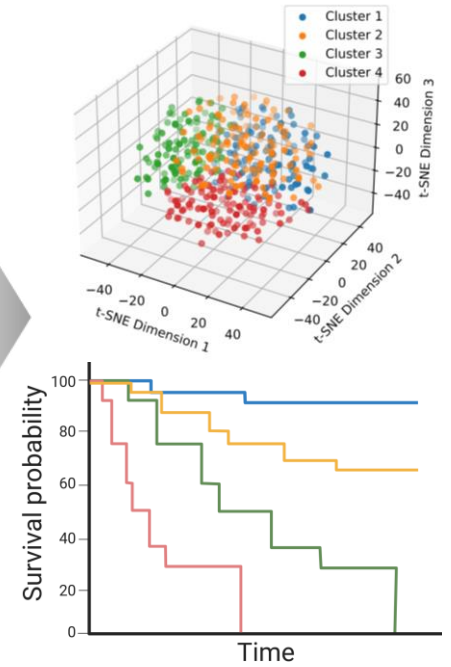
disease network



artificial neural network



patient clustering



In 29 out of 31 cancer types, the clustered patients show significantly different overall survival times

Community profile



Classification

Hierarchical clustering
 $n = \#$ of tumor grade

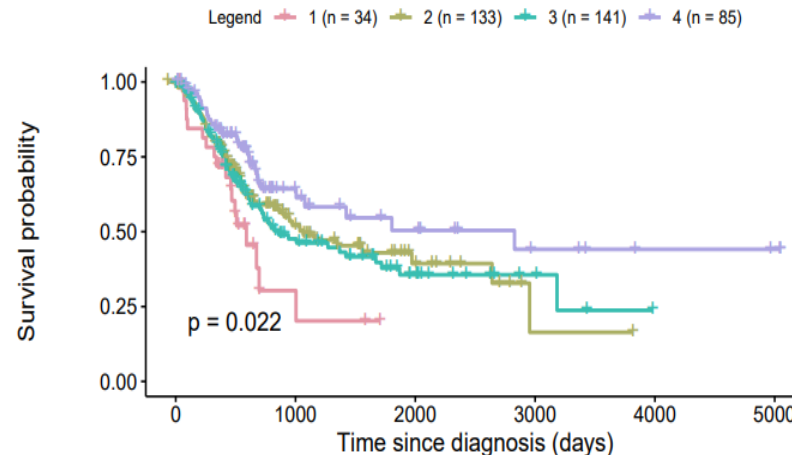
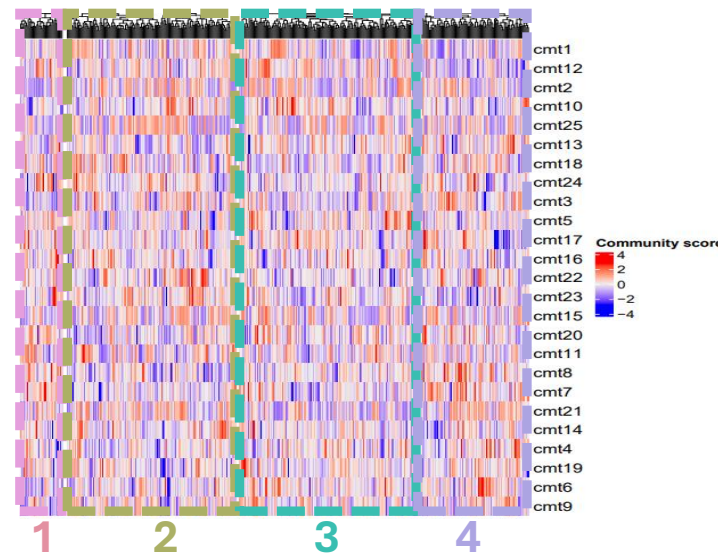


Survival analysis

Kaplan-Meier plot



Bladder Cancer (BLCA)

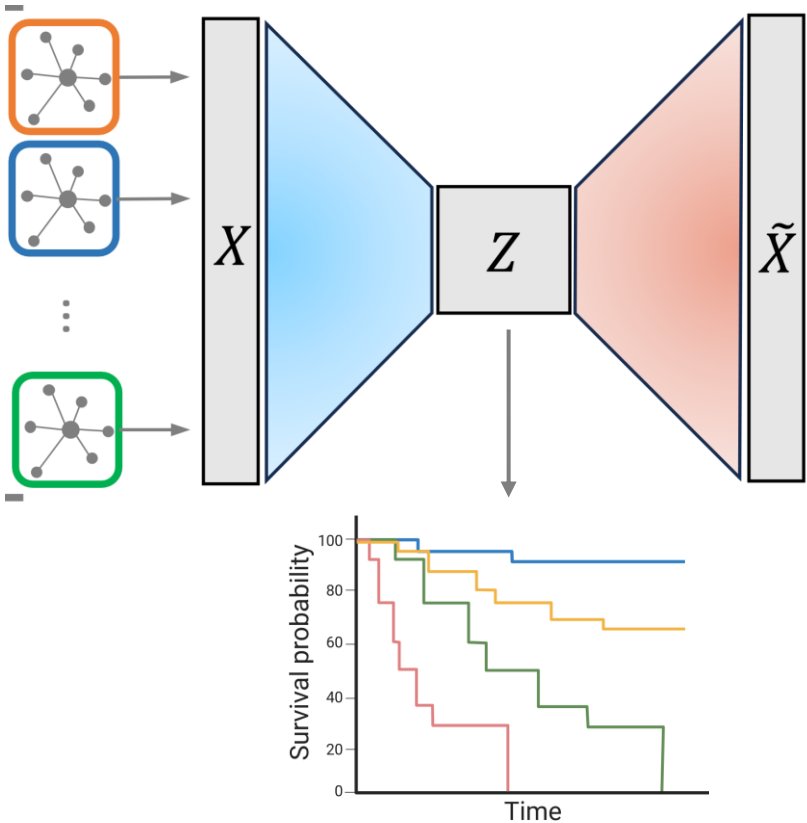


$p \leq 0.05$
(29)

$p > 0.05$
(2)

Cancer name	p-value	Cancer name	p-value
BLCA	0.022	THYM	0.022
BRCA	0.045	UCEC	0.03
COAD	0.0001	TGCT	0.0001
ESCA	0.043	MESO	0.021
KICH	0.015	UVM	0.012
KIRC	0.011	HNSC	0.18
KIRP	0.024	UCS	0.00023
LIHC	0.0001	CHOL	0.024
LUAD	0.049	CESC	0.014
LUSC	0.0001	ACC	0.021
PAAD	0.018	PCPG	0.00091
PRAD	0.023	DLBC	0.0001
READ	0.036	LGG	0.11
SKCM	0.003	SARC	0.0001
STAD	0.0017	GBM	0.0001
THCA	0.0013		

Model interpretation



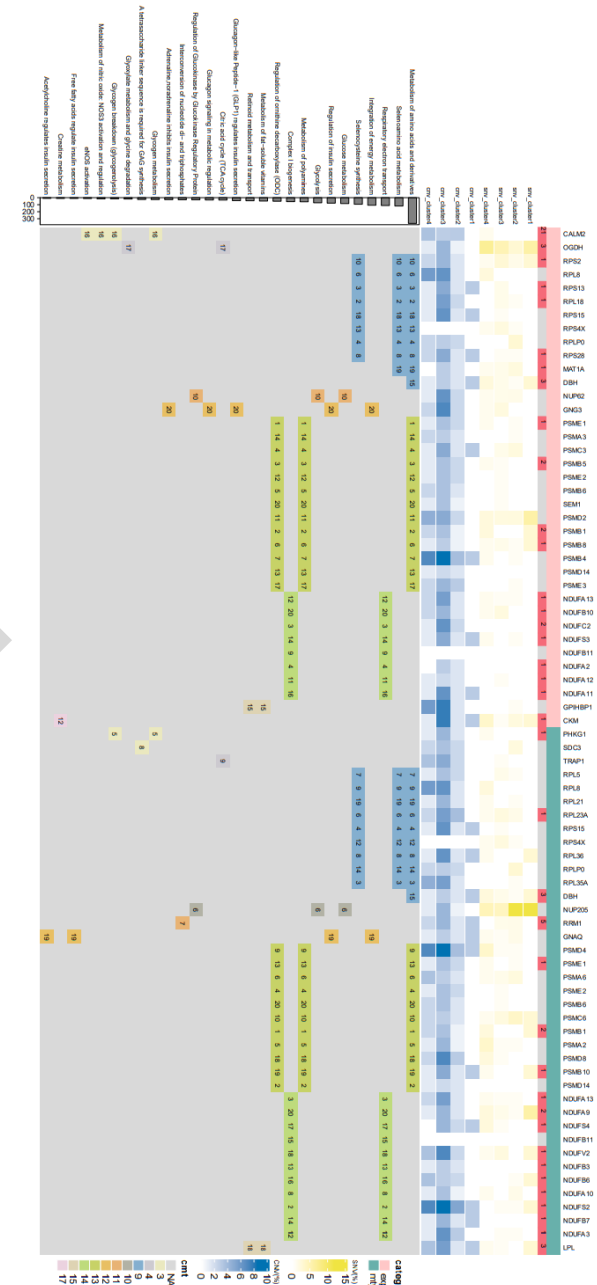
Patients' genomic profiles (SNV and CNV)

SHapley Additive exPlanations (SHAP)

Gene set enrichment analysis (Reactome with 2000+ pathways in 29 categories)

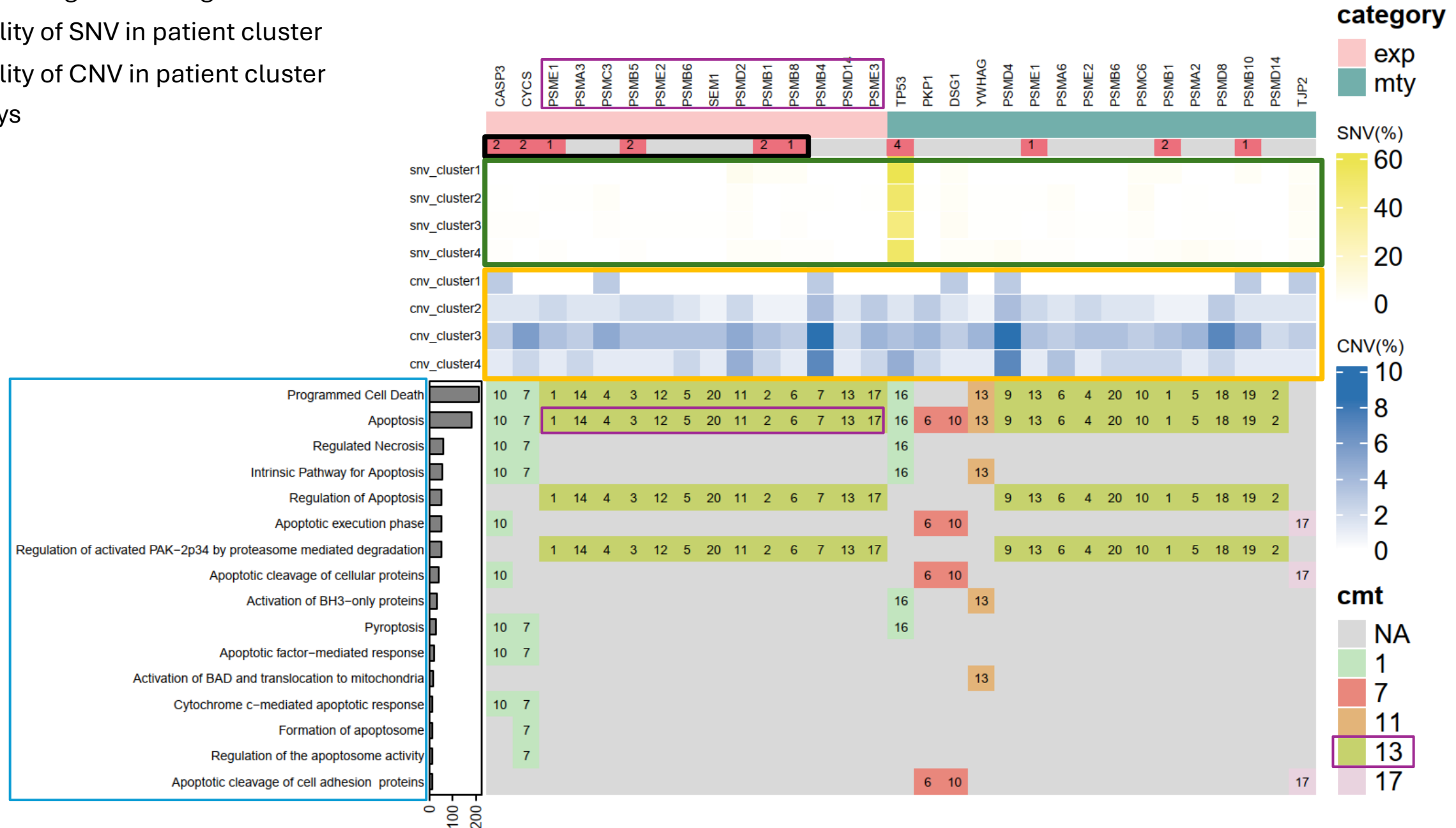
Drug-gene interactions (DrugBank)

SNV: single nucleotide variation
CNV: copy number variation



-  : top-ranking features by SHAP
-  : approved drugs from DrugBank
-  : probability of SNV in patient cluster
-  : probability of CNV in patient cluster
-  : pathways

BLCA - ProgrammedCellDeath



Conclusion

- We propose a systems medicine framework for stratifying patients through integrated analysis of gene expression and DNA methylation signatures.
- Using a novel AI framework informed by cancer networks, we systematically identify distinct patient clusters with divergent survival trajectories across 29 of 31 TCGA cancer types.
- Utilizing the interpretable SHAP method, we quantitatively assess the relative importance of genomic features contributing to patient classification.
- Our approach employs parallel computing infrastructure to efficiently process and analyze large-scale genomic datasets, enabling robust training of advanced AI predictive models.

Acknowledgement

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